

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Embedded Systems

COURSE CODE: ECA1426

COURSE OUTCOME COVERED

CO1: Analyze and design complex and networked embedded systems using formal design methodologies, embedded architectures, and communication protocols (such as CAN and CAN FD), for real-world applications. (BL5)

Assessment Tool : Concept Mapping

Assessment Tool Weightage: 30%

Duration : 30 minutes

Marks: 20

S.NO	QUESTIONS	BTL	MARKS
		BL5	20

Evaluation Rubric (20 Marks per Question – Concept Map)

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Code of Conduct:

I S. PRITHISH (192421134) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Prithish

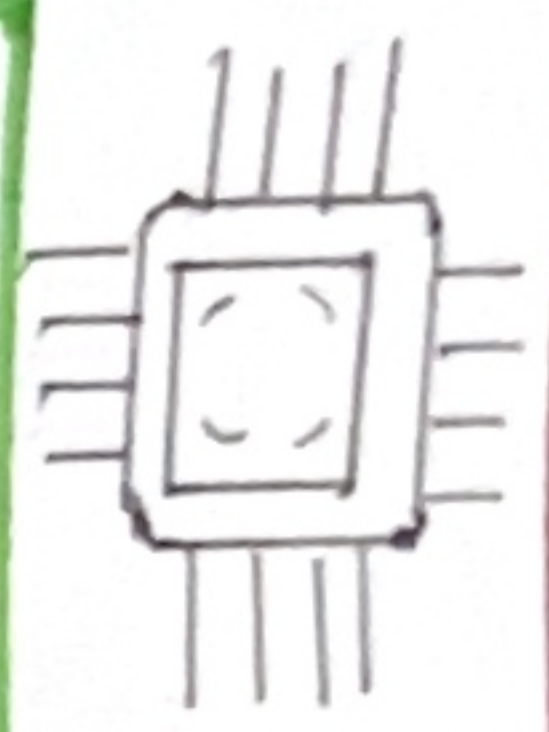
Student Name & Register No.	Concept Identification (5)	Linkages (5)	Organization (4)	Integration of Literature (4)	Presentation (2)	Total (20)
192421134						

ECA-1426

EMBEDDED SYSTEM
COL. ATI

"S. PATHISHAN"
(192421134)

SELECTION OF PROCESSING UNITS



CONCEPT MAP



3

Size/Form

- * Peripheral
- * Scalability
- * Security
- * Ease of Development

Computational Speed

- * Data Handling
- * Algorithmic Complexity

Battery Operated?

- * Power Consumption
- * Energy Efficiency
- * Thermal Limits

Unit cost

- * Development cost
- * Manufacturing cost
- * Maintenance cost

Response Time

- * Deterministic Behaviour
- * Interrupt Handling
- * Latency Requirements

Optimal performance

- * Low power
- * Low cost
- * High Reliability

SELECTION OF PROCESSING

MCU (MICROCONTROLLER UNIT)

KEY FEATURES

- * Low Power consumption
- * Integrated peripherals
- * Real-Time Control

SUITABLE APPLICATION

- * Washing Machine
- * Microwave Oven
- * Electric fan
- * Home Automation

MPU (MICROPROCESSOR UNIT)

KEY FEATURES

- * High processing power
- * Large memory capacity
- * Multi-tasking consumption
- * High power

SUITABLE APPLICATION

- * Smart TV
- * Industrial PC
- * POS Terminals

DSP (DIGITAL SIGNAL PROCESSING)

KEY FEATURES

- * High Speed Signal processing
- * MAC operations optimized
- * Low latency

SUITABLE APPLICATION

- * Audio processing
- * Video Compression
- * Image processing

SOC (SYSTEM ON CHIP)

KEY FEATURES

- * Integration of CPU, DSP, GPU, Memory, I/O
- * High performance
- * Compact size

SUITABLE APPLICATION

- * Smartphones
- * Tablets
- * IoT Gateways
- * Drones

FINAL OUTCOM



High Performance

Low Power



Cost Effective



Real-Time Response



Reliable & secure